

DIGITAL LOGIC AND SYSTEM DESIGN

LAB ASSIGNMENT 4B

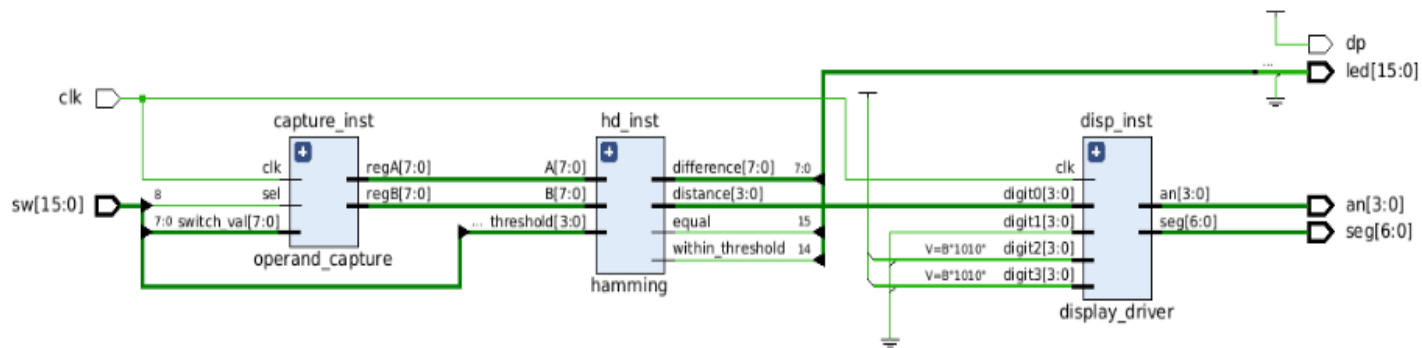
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INTRODUCTION- In this lab we are going to design a circuit for sequentially taking two operands (due to limitations of switches on FPGA board) and a threshold as input and then using hamming file from lab 4a we calculated various outputs and showed them as output in leds and hamming distance on 7seg display . To keep modularity we designed different modules for different subtasks.

For the 7 seg output we have to output a dash (G7 i.e. 7th segment) on first 2 displays then a 0 on 3rd display and hamming distance on 4th display.

BLOCK DIAGRAM



Here the operand capture one part shown above is **capturing registers** then the hamming part shows are **instantiated hamming** block. Then the display driver one is our **display path**.

Here *operandcapture*, *displaydriver* are sequential while *Sevenssegdecoder*, *hamming*, *2to1 8 bit mux*, *hamming_wrapper* are combinational.

FPGA RESOURCE COUNTS -

LUT UTILIZED – 37

FLIP-FLOP UTILIZED – 34

BRAM UTILIZED – 0

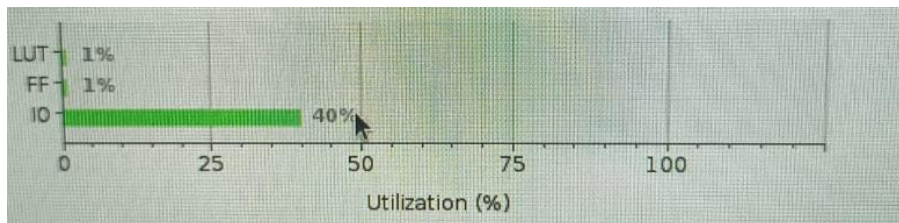
DSP UTILIZED – 0

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	0	0	0	90	0.00

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs*	37	0	0	20800	0.18
LUT as Logic	37	0	0	20800	0.18
LUT as Memory	0	0	0	9600	0.00
Slice Registers	34	0	0	41600	0.08
Register as Flip Flop	34	0	0	41600	0.08
Register as Latch	0	0	0	41600	0.00
F7 Muxes	0	0	0	16300	0.00
F8 Muxes	0	0	0	8150	0.00

2. Memory

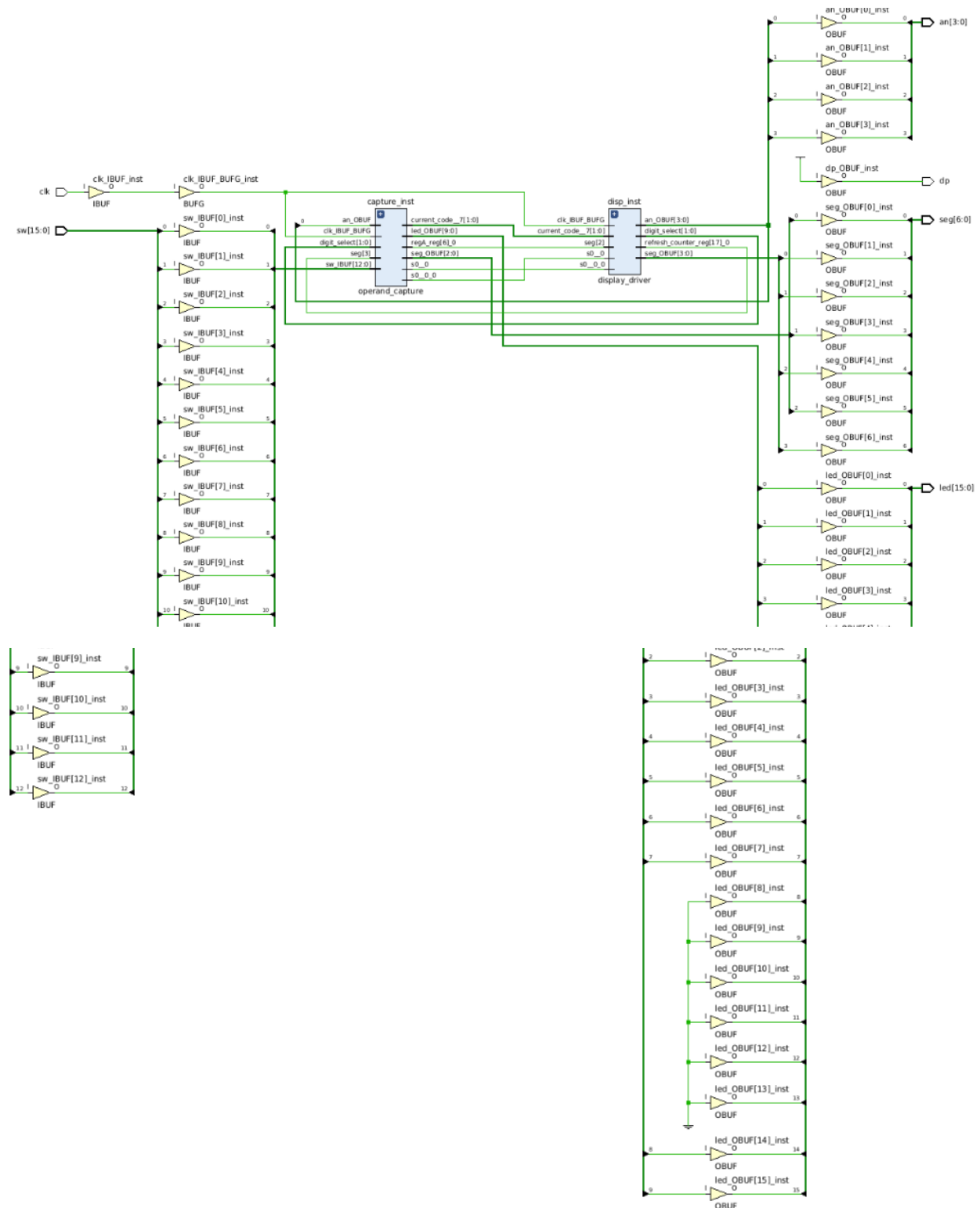
Site Type	Used	Fixed	Prohibited	Available	Util%
Block RAM Tile	0	0	0	50	0.00
RAMB36/FIFO*	0	0	0	50	0.00
RAMB18	0	0	0	100	0.00



Resource	Utilization	Available	Utilization %
LUT	37	20800	0.18
FF	34	41600	0.08
IO	42	106	39.62

UTILIZATION SUMMARY

SYNTHESIZED SCHEMATIC



SCHEMATIC SUMMARY

WE TAKE INPUT FROM THE FPGA BOARD SWITCHES

- SW0-7 FOR BINARY INPUT STRING/COMBINATION-OF-SWITCHES
- SW8 TO SELECT BETWEEN A AND B
- SW 9- 12 FOR THREASHOLD
- Clk mainly for driving all the sequeuntial logic in the code

THE OPERATOR CAPTURE MODULE

- Takes inputs the input binary string/combination-of-switches , sel to indicate which of A/B to calculate (also the clock)
- It assigns A and B their correct value after choosing whom to give the binarystring by using multiplexers
- The first MUX has D0 as sw[7:0] and D1(in1) as A(oldvalue) itself so if sel is 1 then A gets updated to its old value so it doesnt change and if sel is 0 then A get the value of sw[7:0]
- Similarly the second mux assigns D0(in0) to B(oldvalue) and D1 to sw[7:0] so B only gets updated if sel is 1.

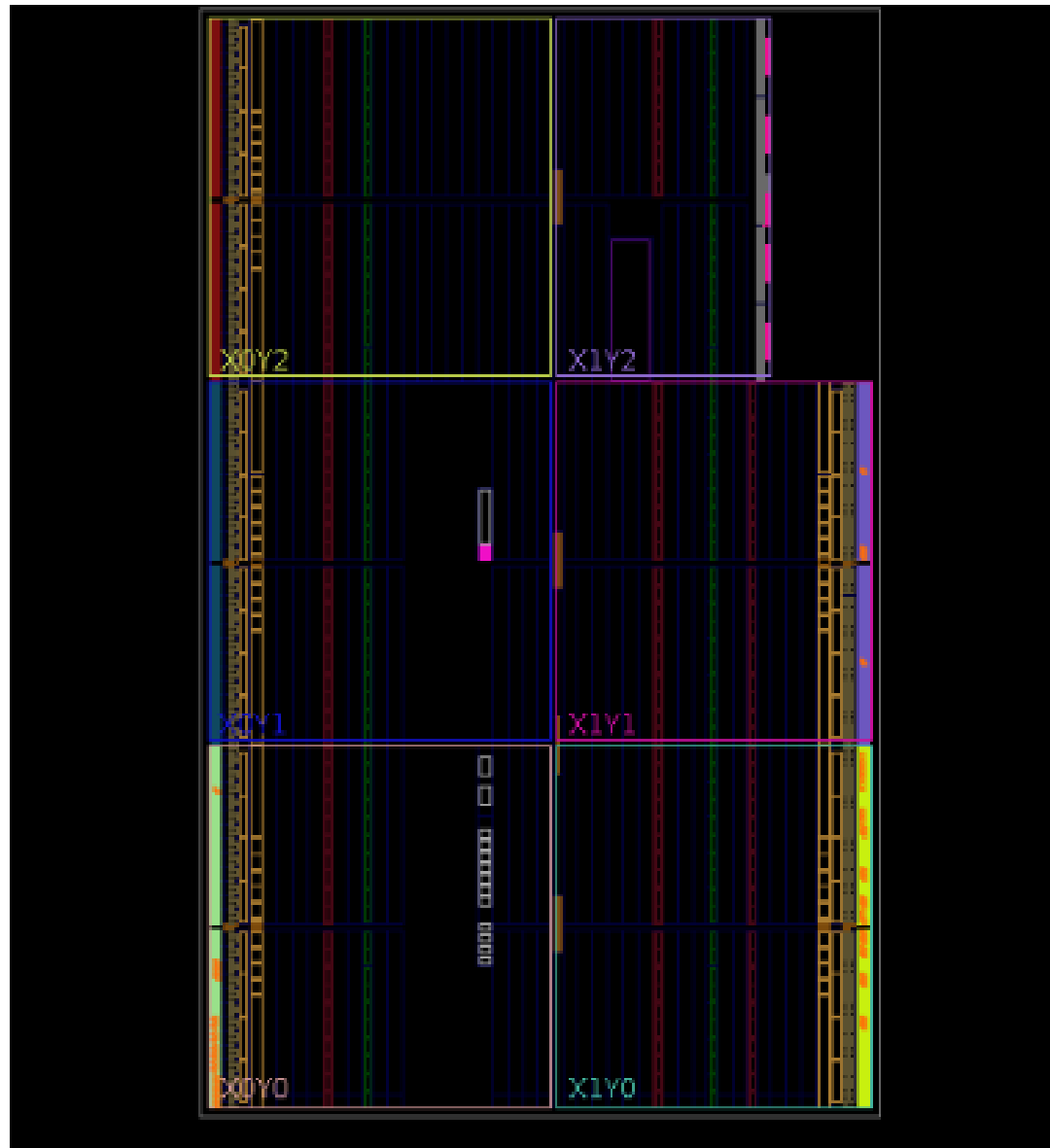
HAMMING

- Takes input the threashold , string A and B and outputs their difference string, distance, whether they are equal or not , whether they are in threshold or not
- Used lab4a modules

DISPLAY DRIVER

- Takes input a clock
- Takes input for all the 4 digits
- Swicthes between displaying just digit 1 ,then just digit 2 , then displaying just digit 3 , then displaying just digit 4 then back to 1 and reapeats at a rapid pace so isnt percivable by human eye

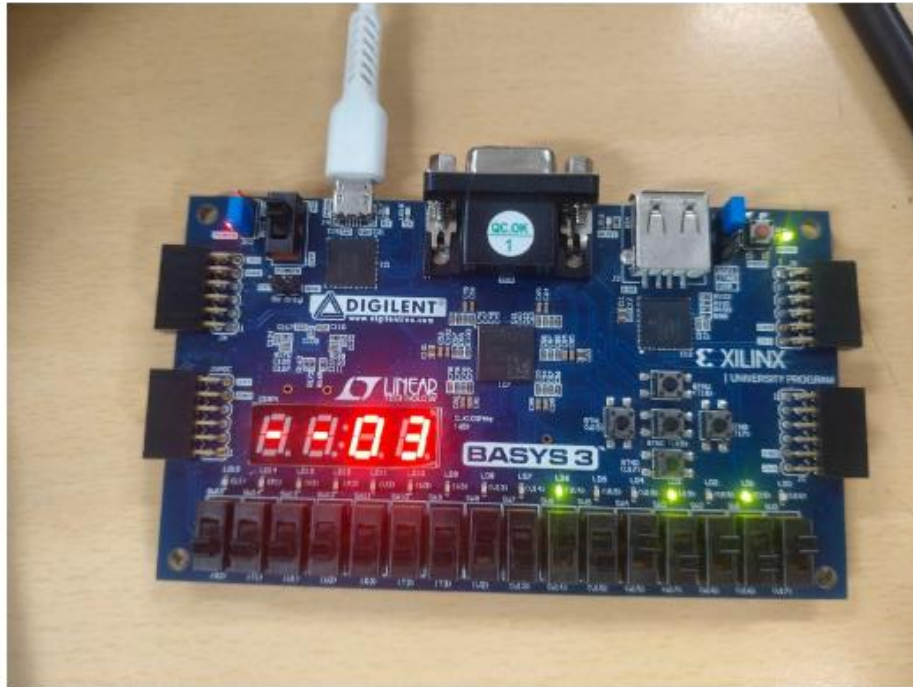
- For each digit i
 - o Anode- i is turned to 0 others to 1 switching a_i on
 - o 7 cathodes are given 0 or 1 based on prespecified encoding



FLOOR PLAN FOR CONSTRAINT FILE

FPGA WORKING

A=00011111,B=01010101,T=0



A=10000001,B=00011110,T=3

